

Alexander Kvist

Department of Neurobiology, Care sciences and Society | Karolinska Institutet

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Education

KTH Royal Institute of Technology

Stockholm, Sweden

MSc in Medical Engineering

2016–2018

Further specialization with courses in medical imaging systems, electronics, implants, informatics, patient safety.

EPFL École Polytechnique Fédérale de Lausanne

Lausanne, Switzerland

Exchange Studies

Sep 2016–Jul 2017

One year of exchange studies at master level. Specialized in data analysis, machine learning, simulation methods and neuroscience.

KTH Royal Institute of Technology

Stockholm, Sweden

BSc in Medical Engineering

2013–2016

Broad engineering base with courses in mathematics, statistics, physics, programming, anatomy and medical imaging systems.

Doctoral education

Karolinska Institutet

Stockholm, Sweden

PhD in Medical Science

2021–2026

My thesis focused on researching gait and balance in neurodegenerative populations, from behavior to neurophysiology. More specifically, the thesis contributed to a better understanding of motor behavior during complex walking, cognitive-motor interference and neural correlates of these, in order to support evaluating and targeting interventions for these abilities.

Work Experience

uMOVE core facility, Karolinska Institutet

Stockholm, Sweden

Lab Manager

Jan 2022–present

Research support, running experiments and analysing data for internal and external clients of the core facility.

Franzén Group, Karolinska Institutet

Stockholm, Sweden

PhD Student

Sep 2021–Mar 2026

PhD Student in medical science.

Sectra Imaging IT

Linköping, Sweden

Software Engineer

Aug 2018–Aug 2021

Software development work on a large PACS server platform for hospital radiology departments.

- Developed high-performance .NET server components for a large PACS platform in C# and C++
- Had a role as agile coach and scrum master

Science for Life Laboratory

Stockholm, Sweden

MSc Degree Project

Jan 2018–Jun 2018

Developed a method for classifying pathogenic mutations in proteins.

- Built a machine learning model based on Deep Learning techniques with state-of-the-art performance in benchmarks
- Worked with large amounts of data, including parallel processing at an HPC2N cluster, and developed an end-to-end pipeline from raw data to results

Center for Molecular Medicine, Karolinska Institutet

Stockholm, Sweden

BSc Degree Project

Mar 2016–May 2016

Developed a visualization tool for structural variants in DNA.

- Created an easy-to-use platform independent GUI in Qt
- Worked with data from whole genome sequencing

Publications

Peer-reviewed articles

- [1] **Kvist, A.**, Peterson, D. S., Bezuidenhout, L., Johansson, H., Albrecht, F., Ekman, U., Conradsson, D. M., and Franzén, E. Jan. 2026. "Exploring Gait Automaticity and Prefrontal Brain Activity during Single and Dual-Task Walking in Aging and Parkinson's Disease". In: *Journal of NeuroEngineering and Rehabilitation*. DOI: 10.1186/s12984-025-01864-w.
- [2] Albrecht, F., **Kvist, A.**, and Franzén, E. Jan. 2025. "Resting-State Functional near-Infrared Spectroscopy in Neurodegenerative Diseases – A Systematic Review". In: *NeuroImage: Clinical* 45, p. 103733. DOI: 10.1016/j.nicl.2025.103733.
- [3] Yücel, M. A., [...], **Kvist, A.**, et al. Aug. 4, 2025. "fNIRS Reproducibility Varies with Data Quality, Analysis Pipelines, and Researcher Experience". In: *Communications Biology* 8.1, p. 1149. DOI: 10.1038/s42003-025-08412-1.
- [4] **Kvist, A.**, Bezuidenhout, L., Johansson, H., Albrecht, F., Moulæe Conradsson, D., and Franzén, E. Jan. 2024. "Validation of fNIRS Measurement of Executive Demand during Walking with and without Dual-Task in Younger and Older Adults and People with Parkinson's Disease". In: *NeuroImage: Clinical* 43, p. 103637. DOI: 10.1016/j.nicl.2024.103637.
- [5] **Kvist, A.**, Tinmark, F., Bezuidenhout, L., Reimeringer, M., Conradsson, D. M., and Franzén, E. Jan. 2024. "Validation of Algorithms for Calculating Spatiotemporal Gait Parameters during Continuous Turning Using Lumbar and Foot Mounted Inertial Measurement Units". In: *Journal of Biomechanics* 162, p. 111907. DOI: 10.1016/j.jbiomech.2023.111907.
- [6] **Kvist, A.**, Bezuidenhout, L., Johansson, H., Albrecht, F., Ekman, U., Conradsson, D. M., and Franzén, E. 2023. "Using Functional Near-Infrared Spectroscopy to Measure Prefrontal Cortex Activity during Dual-Task Walking and Navigated Walking: A Feasibility Study". In: *Brain and Behavior* 13.4, e2948. DOI: 10.1002/brb3.2948.

Preprints and pre-registrations

- [7] **Kvist, A.** and Wallin, A. Jan. 2025. *AS-RTS - Comparison between Auditory Stroop with Pre-Defined Intervals and Auditory Stroop with Response Triggered Stimuli*. DOI: 10.17605/OSF.IO/85HT3.

PhD thesis

- [8] **Kvist, A.** Feb. 4, 2026. "Evaluating Complex Walking in Aging and Neurological Disease : From Motor Behavior to Brain Activity". thesis. Karolinska Institutet. DOI: 10.69622/30801290.v1.

MSc thesis

- [9] **Kvist, A.** 2018. "Identifying Pathogenic Amino Acid Substitutions in Human Proteins Using Deep Learning". MA thesis. KTH, School of Engineering Sciences in Chemistry, Biotechnology, Health (CBH) / KTH, School of Engineering Sciences in Chemistry, Biotechnology, and Health (CBH), p. 33.

Awards

Society for functional near-infrared spectroscopy (SfNIRS) **Birmingham, England**
Poster award **2024**
Poster award for poster "Prefrontal cortex activity during navigated walking in older adults and people with Parkinson's disease" during fNIRS 2024

Funding

Norrbacka-Eugenia foundation **Stockholm, Sweden**
PhD funding **2023-2025**
Funding for PhD project, around 1 million SEK or 100 000 USD

Teaching

uMOVE core facility **Stockholm, Sweden**
Movement analysis workshops **2022-2025**
Teaching and demonstration of analysis of spatiotemporal gait parameters using inertial-based movement sensors in a laboratory environment, as well as with a pressure-sensitive mat. Also demonstration of functional near-infrared spectroscopy (fNIRS) measurements. Leading of students at basic and advanced level in performing their own data collection and interpretation.

Seminars, presentations, posters

Neuroimaging Methods Symposium (NEMES) **Gothenburg, Sweden**
Workshop: NEMES 2025 **2025**
"Intro to fNIRS"

Neuroimaging Methods Symposium (NEMES) **Gothenburg, Sweden**
Lecture: NEMES 2025 **2025**
"Introduction to fNIRS: basic principles and applications"

Neuroförbundet **Stockholm, Sweden**
Lecture: NEURO Stockholm **2025**
"Komplex gång och Parkinsons sjukdom"

International Society for Posture and Gait Research **Maastricht, Netherlands**
Poster: ISPGR World Congress 2025 **2025**
"Exploring gait automaticity and prefrontal brain activity during single and dual-task walking in aging and Parkinson's disease"

Karolinska Institutet **Stockholm, Sweden**
Presentation: Tissue & Motion conference **2024**
"Understanding complex walking in Parkinson's disease, from gait analysis to neuroimaging"

Umeå University **Umeå, Sweden**
Presentation: Psychology department Friday seminar series **2024**
"Neuroimaging complex walking using functional near-infrared spectroscopy - experimental design, methodology and analysis"

Society for functional near-infrared spectroscopy (SfNIRS) **Birmingham, England**
Poster: fNIRS 2024 **2024**
"Prefrontal cortex activity during navigated walking in older adults and people with Parkinson's disease"

Karolinska Institutet **Stockholm, Sweden**
Presentation: Imaging seminar series **2023**
"Understanding complex walking in Parkinson's disease, from gait analysis to neuroimaging" (with Rodrigo Vitorio)

Karolinska Institutet **Stockholm, Sweden**
Presentation: Neurocenter seminar series **2023**
"Movement analysis and brain activity in Parkinson's Disease and Multiple Sclerosis" (with Andreas Wallin)

International Society for Posture and Gait Research

Brisbane, Australia

Poster: ISPGR World Congress 2023

2023

"Validation of fNIRS measurement of cognitive load during complex walking tasks"

International Society for Posture and Gait Research

Montreal, Canada

Poster: ISPGR World Congress 2022

2022

"Validation of algorithm for calculating spatiotemporal gait parameters during continuous turning using a wearable lumbar accelerometer"